Ku-Yu Fan

PhD Candidate in Mathematics | Nagoya University

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RESEARCH INTERESTS

Multiple zeta values; motivic multiple zeta values and coactions; Schur multiple zeta values; arborified multiple zeta values; p-adic multiple zeta values; multiple polylogarithms; p-adic multiple L-functions.

EDUCATION

2024-2027* **PhD in Mathematics, Graduate School of Mathematics, Nagoya University (expected September 2027)**

Advisor: Hidekazu Furusho

2021-2024 **Master of Science (Mathematics), Institute of Mathematics, National Taiwan University (NTU) (Advisor: Nobuo Sato)**

2017-2021 **Bachelor of Science (Mathematics), Department of Mathematics, National Tsing Hua University (NTHU)**

REFEREED RESEARCH ARTICLES

2026 **A map between arborifications of multiple zeta values**

Ku-Yu Fan | Journal of Algebra | arXiv:2508.20387

2026 **Coaction formula for the motivic version of Yamamoto's integral**

Ku-Yu Fan | Mathematische Zeitschrift | Vol. 314, Article 35 | DOI: 10.1007/s00209-026-04138-w | arXiv:2406.08127

PREPRINTS

2026 **Regularised double shuffle relations for planar Arborified Zeta Values**

Pierre Catoire, Pierre J. Clavier, Ku-Yu Fan | arXiv:2609.02346 [math.NT]

2026 **p -adic multiple L -functions and twisted multiple Bernoulli numbers**

Ku-Yu Fan | arXiv:2607.04086 [math.NT]

2026 **p -adic multiple zeta values of integer indices**

Ku-Yu Fan | arXiv:2603.22923 [math.NT]

EXPOSITORY AND OUTREACH PUBLICATIONS

2021 **The Number of Lattice Lines in Lattice Figures [in Chinese]**

Ku-Yu Fan and Feng-Mei Lin | Mathmedia (Shuxue Chuanbo), 45(2), 44-53 | Mathematics exposition (in Chinese)

SELECTED RESEARCH TALKS

2026-09-17
Explicit Formulas for Cyclotomic Multiple Bernoulli Numbers
NCTS Number Theory Seminar | National Tsing Hua University

2026-09-01
An explicit formula for the special values of the p-adic multiple L-functions
MSJ Autumn Meeting 2026 | Kobe University

2026-07-27
Double shuffle relations for p-adic multiple zeta values of integer indices
The 5th International Workshop on Multiple Zeta Values and Related Topics | Central South University

2026-07-07
Double shuffle relations for integer indices and their application to -adic multiple zeta values
25th Sendai-Hiroshima Number Theory Conference | Tohoku University

2026-06-19
p-adic multiple zeta values and double shuffle relations for integer indices
Algebraic and Combinatorial Perspectives in the Mathematical Sciences | Online

2026-03-16
p-adic multiple zeta values of integer indices
Meeting in the Middle: Conference on characteristic 0 and characteristic p multiple zeta values (and other topics) | University of Hawai'i at Manoa

2026-02-17
p-adic multiple L-function and twisted multiple Bernoulli numbers
MATHSCI FRESHMAN SEMINAR 2026 | Nagoya University

2025-09-29
A Map Between Arborifications of Multiple Zeta Values
2025 RIMS Symposia (open) "Various Aspects of Multiple Zeta Values" | Kyoto University

2025-09-16
A Map Between Arborifications of Multiple Zeta Values
MSJ Autumn Meeting 2025 | Nagoya University

2025-07-12
A Map Between Arborifications of Multiple Zeta Values
The 4th International Workshop on Multiple Zeta Values and Related Topics | Tongji University

2024-11-08
Coaction Formula for the Motivic Version of Yamamoto's Integral
2nd Kindai Workshop Multiple Zeta Values and Modular Forms | Kindai University

2024-05-23
Coaction formula for the motivic version of Yamamoto's integral
Monthly Meeting | Online

FELLOWSHIPS AND AWARDS

2024-2027
Japan-Taiwan Exchange Association Scholarship (Graduate Research Students)
Fellowship / scholarship

2019
Professor Han-Ching Lai Memorial Scholarship
Fellowship / scholarship

TEACHING EXPERIENCE

Spring 2026	**Special Topics in Foundations (Analysis)** Teaching Assistant, Nagoya University
Fall 2025	**Calculus** Teaching Assistant, Nagoya University
Spring 2025	**Calculus** Teaching Assistant, Nagoya University
Spring 2024	**Calculus** Teaching Assistant, National Taiwan University
Fall 2023	**Calculus** Teaching Assistant, National Taiwan University
Spring 2023	**Calculus** Teaching Assistant, National Taiwan University
Spring 2023	**Modern Algebra** Teaching Assistant, National Taiwan University
Fall 2022	**Calculus** Teaching Assistant, National Taiwan University
Fall 2022	**Modern Algebra** Teaching Assistant, National Taiwan University
Spring 2022	**Linear Algebra** Teaching Assistant, National Taiwan University
Fall 2021	**Linear Algebra** Teaching Assistant, National Taiwan University
Spring 2021	**Elementary Number Theory** Teaching Assistant, National Tsing Hua University
Spring 2021	**Calculus** Teaching Assistant, National Tsing Hua University
Fall 2020	**Calculus** Teaching Assistant, National Tsing Hua University
Spring 2020	**Advanced Calculus** Teaching Assistant, National Tsing Hua University

ACADEMIC SERVICE AND ORGANIZATION

2026	**Nagoya-Kansai Young Researchers' Zeta Meeting - Organizing Committee (2026; joint 19th Zeta Young Researchers' Meeting / 70th Kansai MZV Seminar; supported by a selected 2025 Nagoya University student project)** Organizing role
2024-	**Monthly Research Seminar (MZV) - Co-organizer (2024-present)** Organizing role

PATENTS IN MATHEMATICS EDUCATION

2021 — **Single Stacking Element Structure and Single Stacking Component Structure for Multi-Style Combinations**

Taiwan utility model M612210 | Co-invented with Wen-Liang Hung

2018 — **Rhombic Dodecahedron Puzzle and Multiple Rhombic Dodecahedron Puzzle**

Taiwan utility model M566100; US patent 10,722,782 | Co-invented with Wen-Liang Hung

2017 — **Hexagonal Prismatic Packing Puzzle**

Taiwan utility model M548578; US patent 10,322,337 | Co-invented with Wen-Liang Hung